

SECTION A – 40 MARKS

(Attempt *all* questions from this *Section*.)

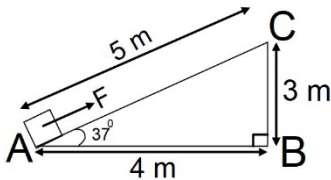
Question 1

[15]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

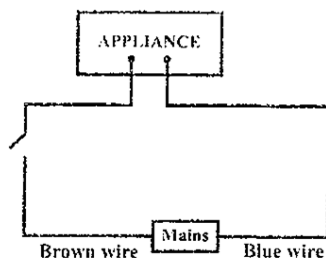
- (i) When a bell is fixed on a cycle rings, then the energy conversion that takes place is:
 - (a) gravitational potential energy to sound energy
 - (b) kinetic energy to sound energy
 - (c) sound energy to electrical energy
 - (d) sound energy to mechanical energy
- (ii) A door lock is opened by turning the lever (handle) of length 0.2 m. If the moment of force produced is 1 Nm, then the minimum force required is:
 - (a) 5 N
 - (b) 10 N
 - (c) 20 N
 - (d) 0.2 N
- (iii) A force '**F**' moves a load from A to C as shown in the figure below. For the calculation of the work done, which of these lengths would you use as the displacement?



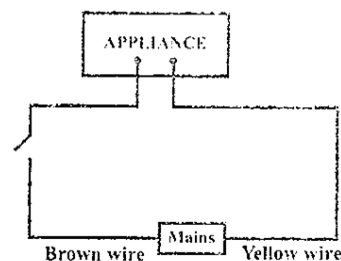
 - (a) 3m
 - (b) 4m
 - (c) 5m
 - (d) 7m
- (iv) A radioactive nucleus containing 128 **nucleons** emits a β – particle. After β – emission the number of **nucleons** present in the nucleus will be:
 - (a) 128
 - (b) 129
 - (c) 124
 - (d) 127
- (v) **Assertion (A):** Ultraviolet radiations are scattered more as compared to the microwave radiations.
Reason (R): Wavelength of ultraviolet radiation is more than the wavelength of microwave radiation.
 - (a) Both A and B are true.
 - (b) A is true but R is false.
 - (c) A is false but R is true
 - (d) Both A and R are false.

- (vi) When the stem of vibrating tuning fork is pressed on a table, the tabletop starts vibrating. These vibrations are **definitely** an example of:
- resonance
 - natural vibrations
 - forced vibrations
 - damped vibrations
- (vii) Which of the following is a class III lever?
- Pair of scissors
 - Wheelbarrow
 - Crowbar
 - Human forearm
- (viii) The specific resistance of a conductor depends on its:
- length
 - material
 - area of cross section
 - radius
- (ix) Identify the option that displays the **correct wiring** with **correct colour code**:

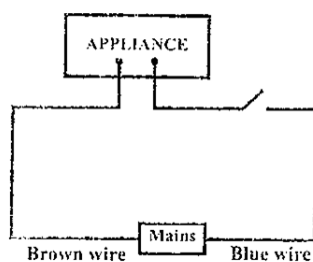
(a)



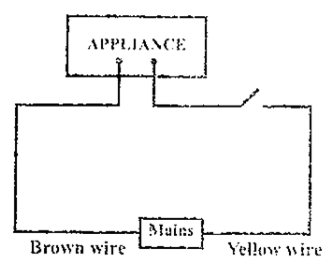
(b)



(c)

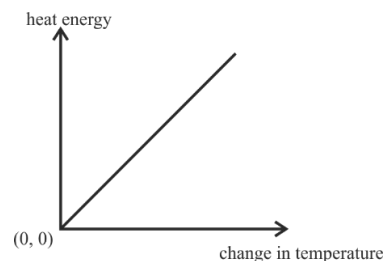


(d)

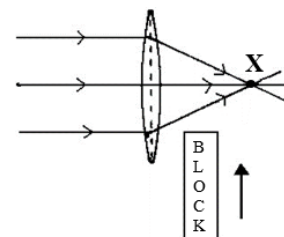


- (x) The potential difference between terminals of a cell in a closed electric circuit is:
- terminal voltage
 - electro motive force
 - voltage drop
 - none of these

- (xi) During melting of ice at 0°C the:
- energy is released and temperature remains constant.
 - energy is absorbed and temperature remains constant.
 - energy is released and temperature decreases.
 - energy is absorbed and temperature increases.
- (xii) Linear magnification(m) produced by a concave lens is:
- $m < 1$
 - $m > 1$
 - $m = 1$
 - $m = 2$
- (xiii) A radioactive element is placed in an evacuated chamber. Then the rate of **radioactive decay** will:
- Decrease
 - Increase
 - Remain unchanged
 - Depend on the surrounding temperature
- (xiv) The graph given below shows heat energy supplied against change in temperature when no energy is lost to the surrounding. The slope of this graph will give:
- Specific heat capacity
 - Latent heat of fusion
 - Latent heat of vaporization
 - Heat capacity



- (xv) A block of glass is pushed into the path of the light as shown below. Then the converging point **X** will:
- Move away from the slab
 - Move towards the slab
 - Not shift
 - Move towards the left side of the lens



Comments of Examiners		Suggestions for teachers
(i)	- Most of the candidates answered it correctly. - Some candidates confused gravitational potential energy to sound energy, thinking that the bell is fixed at a height. - Some candidates also wrote electrical energy to sound energy, may be because of rote learning. The doorbell conversion is electrical to sound.	- Discourage rote learning. - While testing students, discuss in class different examples that lead to the same energy conversions. - Explain the difference between gravitational potential energy, elastic potential energy and kinetic energy.
(ii)	- Most candidates got this question correct. - Some of them lost marks because of errors where they multiplied 1 Nm and 0.2 m instead of dividing. - Some candidates were not clear with the concept and answered by guesswork.	- Encourage students to pay attention to units and sign conventions. - Insist on writing the formula and then substitution. - Give more practice on numericals.
(iii)	- Many candidates could not perform well in this question as the basic concepts were not clear. - For selecting the displacement, the direction of force was not taken into consideration. That is why many candidates selected the height 3 m instead of length of slant surface. - However, many students got the answer correct.	- While doing numericals in the class, it is necessary to discuss with the students as to why we are selecting a particular displacement even if it is not mentioned in the question. - Reinforce that the displacement should be in the direction of the force.
(iv)	- Many candidates who were clear with what happens inside the nucleus during beta emission got it correct. - Many candidates who depend on rote learning added 1, as atomic number increases by 1. - Additionally, some wrote 127 as they were confused whether to add 1 or subtract 1.	It is important to make it clear to the students that in order to balance the number between protons and neutrons, a neutron splits into a proton and an electron, keeping the total number of nucleons the same.
(v)	- Most of the candidates answered it correctly. - Common options selected by many candidates were a) A and R are true and c) A is false but R is true.	Give practise to the students to use mnemonics to remember the order of radiations. For e.g.: [Go: gamma, X; x rays, U: ultraviolet, View: visible, Incredible: infra-red, Marathon: microwaves, Run: Radio

	- Probably the topic was not discussed with the candidates as it was not there in the syllabus for quite some time. - Many candidates could not clearly show the relation between scattering, intensity and wavelength. - Radiations in the ascending or descending order was not written.	waves. Go X U View Incredible Marathon Run] Give ample practise to students on Assertion and Reason type questions.
(vi)	- Most of the candidates answered it correctly. - Some candidates confused it with resonance or damped vibrations. - Difference between resonance and forced vibration as well as difference between forced vibrations and damped vibrations is not clear.	Explain in detail to students that: - when one vibrating body is brought in contact with another vibrating body then it always vibrates, but it will vibrate with maximum amplitude only in case of resonance. - during damped vibrations, amplitude only decreases but during forced vibration amplitude can increase or decrease.
(vii)	- A very straightforward question and almost all answered it correctly. - A few candidates answered it incorrectly due to the concept being not clear however some of them answered correctly but supported it with the wrong diagram which was not asked.	- Advise students to make rough sketches to identify the position of L, F and E. - Insist on identifying the lever by locating the position of L, F and E rather than simply rote learning. - Make students aware that to carry out the same purpose there can be levers belonging to the different classes of lever and in such cases, by supporting the answer with a sketch of the lever.
(viii)	Most candidates answered this question correctly but some were not clear with the concept of specific resistance as they confused it with the factors affecting the resistance.	Use an analogy in class to teach the concept of resistance. E.g. density can be compared with specific resistance as the density depends only on material and not on mass or volume of substance, same way specific resistance depends on material and not on length or cross-sectional area etc.
(ix)	- Many candidates answered it correctly. - Some candidates could not relate colour coding with L, N and E wires. - Some candidates found it difficult to apply both conditions i.e. correct position of switch and also correct wires used.	- Make students aware that in attempting MCQ's, they need to verify all options even if they feel the initial option is correct. Options can be deceptive, and all may appear plausible. - Train students in drawing circuit diagram with the correct position of switch, Fuse, live, neutral and earth wire.

		- Advisable to teach this concept with diagrams, models and videos from authentic source.
(x)	• Even though many candidates answered it correctly, it was clear that the concept of terminal voltage and e.m.f. was not clear to them. They missed out the keyword in the question 'closed' hence answered incorrectly as e.m.f.	- Explain to the students that e.m.f., terminal voltage and potential drop, all are potential differences but across different resistors. - Use the analogy that in class there are many boys (potential difference) but the name of each boy is different (e.m.f., terminal voltage, potential drop) - Use practical demonstrations for explanation of concepts.
(xi)	• Candidates applied simple logic that when the heat is absorbed the temperature increases but when it is released the temperature decreases. • Latent heat concept is not clear. Not clear whether 0 °C ice absorbs or releases latent heat. • How ever quite a number of candidates answered it correctly.	The following concepts should be made clear to the students: - Heat is a measure of kinetic and potential energy possessed by the molecules of the body. - Heat absorbed/released during rise/fall in temperature affects average kinetic energy of the molecules. - Heat absorbed/released during change of phase never affects average Kinetic Energy (k.e) of the molecules but affects potential energy hence temperature is not affected.
(xii)	• Many candidates answered this question correctly. They found this question quite direct. • However, some candidates answered it wrongly because they were not clear with mathematical symbols greater than and less than or could not correlate magnification as ratio of $I:O$ or $O:I$	- Explain in detail the concept of magnification. - Concept can be practically demonstrated by showing the images formed by concave and convex lenses. May include an assignment to calculate magnification during the formation of real images of convex lenses.
(xiii)	• Concept of effect of physical and chemical changes on radioactive reaction was not clear to some candidates • Candidates mistakenly associated radioactive reactions with the chemical reactions, where there is influence of the physical environment.	- Clearly explain the concept of radioactivity to the students. - Emphasise that radioactivity is independent of all physical and chemical changes.

(xiv)	- Some of the candidates selected options 'a' and 'c'. - Candidates could not connect the concept of slope to different situations.	- Revise the concept of finding the values of physical quantities from the slope of graph (done at length in grade IX). - Calculation of slope in different contexts could be tested in class tests. e.g. slope of latent heat against mass gives specific latent heat, slope of kinetic energy against mass gives $\frac{v^2}{2}$ etc.
(xv)	- Majority of the candidates found this question very confusing to understand making it clear that the candidates are not striking depth in their thinking. - Only some candidates answered it correctly.	- Pushing the block of glass into the path of light is similar to placing the lens in water. - Create similar types of critical thinking questions for regular class tests.

MARKING SCHEME

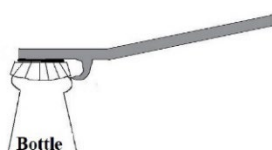
Question 1

(i)	(b) kinetic to sound energy
(ii)	(a) 5 N
(iii)	(c) 5m
(iv)	(a) 128
(v)	(b) A is true but R is false
(vi)	(c) forced vibrations
(vii)	(d) Human forearm
(viii)	(b) material
(ix)	(a)
(x)	(a) terminal voltage
(xi)	(b) energy is absorbed and temperature remains constant.
(xii)	(a) $m < 1$

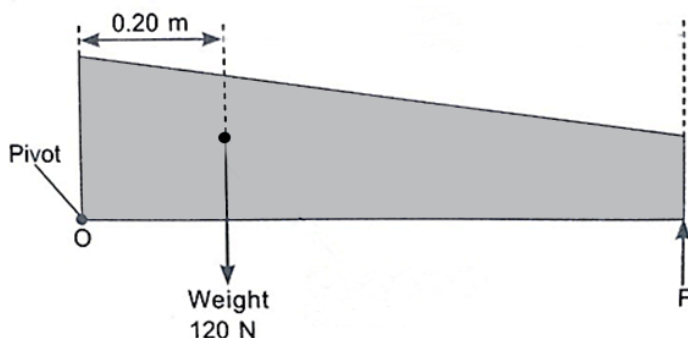
(xiii)	(c) Remain unchanged
(xiv)	(d) Heat capacity
(xv)	(a) Move away from the slab

Question 2

- (i) (a) In the following atoms, which one is a radioisotope? Give *one* use of this isotope. [3]
 O^{16} , C^{14} , N^{14} , He^4
- (b) Name the class of the lever shown in the picture below:



- (ii) Fill in the blanks: [2]
- (a) When a stone tied to a string is rotated in a horizontal plane, the tension in the string provides _____ force necessary for circular motion.
- (b) Work done by this force at any instant is _____.
- (iii) A non uniform beam of weight 120 N pivoted at one end is shown in the diagram [2]
 below. Calculate the value of F to keep the beam in equilibrium.



- (iv) Meera chose to use a block and tackle system of '9' pulleys instead of a single movable pulley to lift a heavy load. [2]
- (a) What is the advantage of using a block and tackle system over a single movable pulley?
- (b) Why should she connect more number of pulleys in the upper fixed block?
- (v) Sumit does 600 J of work in 10 min and Amit does 300 J of work in 20 min. [2]
 Calculate the ratio of the powers delivered by them.
- (vi) 5 bulbs are connected in **series** in a room. One bulb is fused. It is removed and remaining 4 bulbs are again connected in **series** to the same circuit. What will be the effect on the following physical quantities? (Increases, Decreases, Remain Same) [2]

- (a) Resistance
- (b) Intensity of light
- (vii) Rohan conducted experiments on echo in different media. He observed that a [2]
 minimum distance of 'x' meters is required for the echo to be heard in oxygen and 'y'
 meters in benzene. Compare 'x' and 'y'. Justify your answer.
 Speed of sound in oxygen: 340 ms^{-1}
 Speed of sound in benzene: 200 ms^{-1}

Comments of Examiners		Suggestions for teachers
(i) (a)	- Most of the candidates answered this question correctly, but few wrote other options too like He 4, O16. - Several candidates wrote a generalised answer about the use of radioisotope instead of being specific to C14.	- Uses pertaining to the radiations must be discussed in the class. - Make the students aware about the dependent type questions and how these questions are evaluated. - Group discussions and case studies to encourage critical thinking and deeper understanding of the varied uses of radioisotopes in modern society.
(i) (b)	- Most of the candidates identified it correctly as Class II. - Some of them did write it as Class I or Class III as they were not clear how to identify the position of the fulcrum.	- Lay emphasis on identifying the position of F,L and E and avoid rote learning among students. - Bring different types of levers used at home to explain the terms load, fulcrum, effort and also when classify them as first order, second order and third order. - Images from various relevant sources can be used to explain working of levers to students.
(ii) (a)	- Most of the candidates answered it correctly. - Some candidates mentioned rotational, tension force. The concept that centripetal force is required for a body to move in a circular path was not clear to the candidates.	- Explain the concept of centripetal force along with the diagram and all the relevant examples with the help of a mind map. - Demonstrate the concept using demo videos.
(i) (b)	- Most of the candidates answered it correctly. - However, some candidates did write constant instead of zero or wrote formula $W = F \times s$ or some left it blank. - The general experience was, that even though the answer zero is correct for many candidates, the reason was not clear. They thought that on a circular track the	Explain with the examples of positive, negative and zero work clearly indicating the direction of force and displacement.

	displacement was zero, therefore work is zero but they need to understand that the displacement was zero only when the body comes back to its initial position but at every instant the centripetal force is perpendicular to the displacement.	
(iii)	- Most of the candidates were confused with this question. Candidates erroneously considered the anticlockwise moment to be $(Fx0.6)$ instead of $(Fx0.8)$, thus arriving at a wrong answer for the value of 'F' as candidates are at times confused between the concept of fulcrum and C.G. - Majority of the candidates could not interpret the diagram correctly. Only few of the candidates were able to answer it correctly.	- Give rigorous practice on non-uniform distribution of mass. - While solving numerical based on principle of moments, emphasis must be laid on the fact that the fulcrum/pivot is not always placed at the C.G. to balance a body. - When solving problems involving the principle of moments, encourage students to sketch the diagram themselves, clearly indicating the pivot, centre of gravity, and forces to prevent mistakes. Additionally, marking the distances required for calculations directly on the diagram can help mitigate errors effectively.
(iv) (a)	- Many of the candidates answered the first part of the question correctly. - Candidates mentioned the advantages as convenience in changing the direction of effort or can use their weight to lift the object. - Candidates were unable to understand that with an increased number of pulleys, one can have convenient as well as inconvenient direction.	- Dedicate substantial time in explaining how V.R. changes with the increase in the number of pulleys with convenient as well as inconvenient direction. - Encourage students to think of the larger implications of knowledge, rather than just acquiring textbook information. - Actual demonstration can be used to explain the concept.
(iv) (b)	Many candidates failed to comprehend the information given in this part of the question and just repeated the same answer which was given in part (a) of this question. They were unable to understand that the comparison was not between single pulley and the system with 9 pulleys but it was between the number of pulleys in the fixed and movable block of the same system with 9 pulleys.	- Encourage students to grasp the formula for linking efficiency with the number of pulleys in the block and tackle system. It will enhance their understanding of why more pulleys are connected to the upper block. - Emphasis to be laid on the concept that V.R is always equal to the number of strands supporting the load and will be equal to the number of pulleys if the direction of effort is convenient. Also, it will be equal to M.A. only in an ideal machine.
(v)	- Most of the candidates were unable to convert the ratio in the simplest form. - Many candidates did arrive at the final answer in fractional form but	Reiterate to the students that 'compare' questions have to be answered in ratio form i.e. as x:y.

	did not express it in the ratio form i.e. 4:1. - While finding the ratio, many candidates multiplied instead of dividing. - A number of candidates gave the ratio in the form 1:4 instead of 4:1.	Instruct the students to be careful in expressing the ratio in the same order as given in the question.
(vi) (a)	- Most of the candidates answered this question correctly. - Though the question categorically mentioned to answer in terms of 'increases', 'decreases' or 'remains same', candidates used a variety of words to express themselves. - Some candidates made errors by simply assuming one of the given answers without logically deducing it.	- Encourage students to represent circuits diagrammatically, when in doubt as it helps in concept clarity. - Characteristics of series and parallel combinations need to be stressed upon.
(vi) (b)	- Most of the candidates answered this question correctly. - Some candidates were not clear about the relation between resistance, current and intensity/brightness of the bulb glowing. The correct approach is to recognise that when switching from 5 bulbs to 4 bulbs, the resistance decreases, leading to an increase in the intensity.	- Encourage students to always relate the question to a provided equation or diagram before answering. This will help students in minimising errors. - For same P.D; resistance and current are inversely proportional. - Emphasise that the current through the bulb and intensity are directly proportional.
(vii)	- Even though, many of the candidates understood the question, they could not write the steps of the solution correctly. - Most of the candidates were able to arrive at some relation between x and y and state the reason. However, the final answer was not in the expected ratio form (the question was a 'compare' question). - Some candidates did elaborate calculations to arrive at the answer, while others answered logically. - Many candidates did not take time to hear an echo as 0.1s, but were able to consider the same time in both the situations. Some candidates just divided 340 by 200 i.e. obtained 17:10 but just randomly. Some gave the answer in the form 17x:10y	- Reiterate to the students that 'compare' questions have to be answered in ratio form i.e. as x:y. - Discuss with the students to approach connected questions by logically deriving conclusions using formulas and mathematical calculations, especially when data is provided. - Make sure while discussing that when calculating a ratio, it is not necessary to have every value provided initially in the question; initiating it on both ends allows the values to cancel out. - Give rigorous practice in numericals.

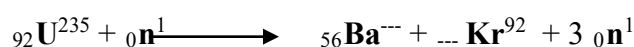
MARKING SCHEME

Question 2

(i)	(a) C^{14} Carbon Dating or implied/ archaeology/ tracers/any other correct use. (b) II/Class II/Type II/ second class/ 2.
(ii)	(a) Centripetal/centre seeking (b) Zero/ 0 /no work.
(iii)	By Pr. of moments $120 \times 0.2 = F \times 0.8 \text{ [Substitution in any form.]}$ $\therefore F = \frac{120 \times 0.2}{0.8} = \frac{120}{4} = 30 \text{ N}$ By Pr. of moments $120 \times 0.2 = F \times (0.8 + x) \text{ [Substitution in any form.]}$ $\therefore F = \frac{120 \times 0.2}{(0.8 + x)} = \frac{120 \times 0.2}{0.8 + x} (N)$
(iv)	(a) (Block and tackle system) increases the M.A.or VR/reduces effort/multiplies force or effort/ to have $MA > 2$ or Implied. (b) To minimise the energy loss (in lifting the movable part of the machine)/ increase the efficiency/ change the effort to a convenient or downward direction/decrease the weight of the lower block/ or implied.
(v)	$P_1 : P_2 = \frac{600}{10 \times (60)} : \frac{300}{20 \times (60)} \text{ / can be explained in the words the mathematical process.}$ $P_1 : P_2 = 4 : 1 \text{ /Explained in word./if 1:4 is written then } P_2 : P_1 \text{ must be mentioned.}$
(vi)	(a) Decreases/reduces/becomes less or implied. (b) Increases/becomes brighter/becomes more or implied.
(vii)	$x > y$ Speed of sound in oxygen > speed of sound in benzene/or implied. Or $x : y = \frac{340 \times 0.1}{2} : \frac{200 \times 0.1}{2} \text{ [time taken is 0.1 s]}$ $x : y = 17 : 10$

Question 3

- (i) (a) In a reading glass what is the position of the object with respect to the convex lens used? [2]
- (b) Why can we **not** use concave lens for the same purpose?
- (ii) A fuse is rated 5 A. Can it be used with a geyser rated 1540 W, 220 V. Write **Yes** or **No**. Give supporting calculations to justify your answer. [2]
- (iii) State *two* factors affecting the speed of rotation of the coil in a D.C. motor. [2]
- (iv) How much heat is required to convert 500 g of ice at 0°C to water at 0°C? The latent heat of fusion of ice is 330 Jg⁻¹. [2]
- (v) Copy and complete the nuclear reaction by filling in the blanks. [2]



Comments of Examiners		Suggestions for teachers
(i) (a)	- Many candidates were not able to give the exact position of the object. - Some of the candidates could not relate to the fact that a reading glass is a magnifying glass. Guesswork was used to arrive at the answer.	- Demonstrate the concepts in the class by showing it to the students. - While discussing the position, nature and size of the image formed by the lens, students could be asked to observe the same in their daily life when objects are at different positions from the reading glass. - Adequate practice of drawing diagrams and independently emphasising construction of tables to understand the characteristics of the image and applications will aid in error reduction.
(i) (b)	- Many candidates answered this question correctly. - Some candidates related this concept to the divergent nature of the lens instead of size. - Few of the candidates failed to understand the link with the part (a) of the question.	Emphasise to the students in class that both lenses form virtual images, but the virtual image formed by a convex lens is magnified and by a concave lens it is diminished.
(ii)	- Most of the candidates answered this part of the question correctly. - Even though numerical part was right, due to a lack of understanding of fuse function, certain candidates lost marks. Certain candidates had a misconception that the fuse rating should be lower than the maximum current in order to protect the appliance.	- Explain to the students that current in the fuse can exceed the maximum current but this situation does not remain for a long time due to high resistivity and low melting point of the fuse wire. Appliances have the ability to withstand a large current for a very short duration. - Students can be informed that they are not obliged to answer connected

	- Many candidates made calculation errors and missed mentioning the unit 'ampere'. - The explanation that 'fuse cannot be used' was not supported by any calculation of safe limit of current. Some candidates even did not give a reason. - Some candidates calculated R instead of I.	questions in the sequence presented; instead, they can opt to perform calculations first and then derive the answer.
(iii)	- Most of the candidates got this question correct. - Some candidates did not understand the difference between the turns of the coil and the number of coils. - Many candidates missed writing 'the <u>strength</u> of the magnetic field' and just mentioned magnet. - Few of the candidates provided correct factors but also included incorrect additional ones, resulting in loss of marks. - Induced current, area of cross section were also among the wrong answers as candidates were confused between motor and generator. - It was also observed that in some cases, candidates were not aware of these topics at all.	- Insist that the students write only the appropriate number of factors. - Explain the difference between the number of coils and number of turns in the coil. - Make the students understand through discussion the relation between the factors affecting the force and speed as well as how the couple acting is responsible for the speed of rotation. - Make sure to be aware of the updated physics syllabus and avoid incorrect interpretation.
(iv)	- Most of the candidates answered it correctly as the question was direct and simple. However, a few errors observed were: ➤ Missing unit or writing incorrect unit. ➤ Converting mass to kg but using the same value of latent heat. ➤ Incorrect calculation. ➤ Using the value of latent heat of ice as 336 Jg^{-1} instead of 330 Jg^{-1} that was mentioned in the question.	Emphasise the use of the correct data from the question and mention the unit in the answers.
(v)	- Most candidates got the mass number of 'Ba' incorrect as they answered 143 instead of 141. - Some candidates did not take into consideration the 3 neutrons in the reaction. Since the charge of a neutron is 0, they got the value of atomic number of 'Kr' incorrect.	Lay emphasis on explaining how the nuclear reactions are balanced.

MARKING SCHEME

Question 3

(i)	a. Within or shorter than the focal length/between F and O/ between lens and focal plane/ accept diagram (only check for the object position) or implied. b. (Concave lens gives upright but) diminished image/ since $m < 1$ or implied.
(ii)	$I = \frac{P}{V} = \frac{1540}{220} = 7A \text{ (must)}$ No, it cannot be used.
(iii)	Directly proportional to/ Increase with the increase in) - the current (through the coil). - the strength of the magnetic field/magnet. - the number of turns (of the coil) - the area/width (of the coil). [Any two points.]
(iv)	$H = ml \quad \therefore H = 500 \times 330$ $\therefore H = 165000J$
(v)	${}_{92}\text{U}^{235} + {}_0\text{n}^1 \rightarrow {}_{56}\text{Ba}^{141} + {}_{36}\text{Kr}^{92} + 3 {}_0\text{n}^1$

SECTION B – 40 MARKS

(Attempt **any four** questions from this **Section**.)

Question 4

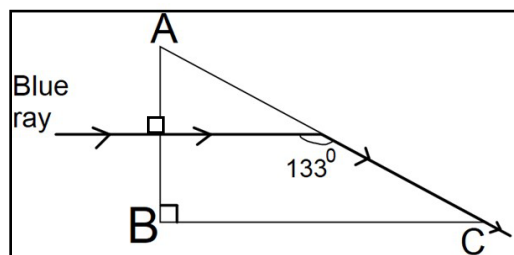
- (i) The image of a **candle flame** placed at a distance of 36 cm from a spherical lens, is formed on a screen placed at a distance of 72 cm from the lens. Calculate the focal length of the lens and its power. [3]

- (ii) Below is an incomplete table showing the arrangement of **electromagnetic spectrum** in the increasing order of their wavelength. Complete the table. [3]

Gamma ray	X – ray	U V rays	Visible rays	Infrared	A	Radio waves
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- (a) Identify the radiation **A**.
- (b) Name the radiation used to detect fracture in bones.
- (c) Name *one* property common to both **A** and Radio waves.
- (iii) (a) Why do we use red colour as a danger signal on the top of a skyscraper? [4]
- (b) The diagram below shows the path of a blue ray through the prism:

- Calculate the critical angle of the material of the prism for blue colour.
- What is the measure of the angle of this prism (A)?
- Which colour should replace the blue ray, for the ray to undergo Total Internal Reflection?



Comments of Examiners		Suggestions for teachers
(i)	- Most of the candidates attempted this question. - The following errors were observed in solving this question: ➤ Interchanged object and image distance. ➤ Used wrong sign convention. ➤ Used wrong lens formula. ➤ Left the answer as a fraction. ➤ Did not convert focal length in 'm' while calculating power.	- Explain sign convention clearly to the students in detail. - Insist on applying sign convention at the time of substitution and not before. - May be classify this kind of numerical into three types. ➤ Real image of a convex lens. [Magnified, diminished and same size] ➤ The virtual image of a convex lens. ➤ The virtual image of a concave lens. Sign conventions are different only for these cases. - Give sufficient practice of these numerical to the students and guide them to write the final answer in decimal. - Show a live demonstration in the class that can help the students understand the position of the object and the image. Teach the sign convention with the help of it.
(ii) (a)	- Most of the candidates answered this question correctly. - A few candidates applied guesswork.	Give practise of similar kind of questions regularly.
(ii) (b)	Almost all candidates identified this radiation correctly.	Give practise of similar kind of questions regularly.
(ii) (c)	- A large number of candidates answered them correctly but some of them due to a lack of revision could not state the common properties. - Some of them gave vague answers such as long or short wavelength etc. - Several candidates wrote 'uses' instead 'of properties'.	- Encourage students to read the content thoroughly and mindfully. - Explain to the students common as well as distinguishing properties of all radiations in the electromagnetic spectrum.
(iii) (a)	- Many candidates answered this question correctly. - Accuracy was lacking in the answer. They identified longer wavelength of red light but instead of relating it to the scattering, they related it to the refraction or just gave a general statement that it can be seen clearly from far.	- Provide clarity to the students in the concepts of scattering, deviation, refraction etc through detailed discussions. - Check the revised syllabus from time to time. - Discuss repeatedly the way to differentiate the applications of refraction and scattering, particularly emphasising the concept of

	Some of the candidates were not aware that the topic is reincluded in the syllabus.	scattering in relation to wavelengths and its conditions.
(iii) (b) 1.	- Even though, many candidates answered the question correctly, it was also observed that some candidates: ➤ were confused. ➤ could not show proper working. ➤ gave angle as 47^0 or 133^0	- Provide rigorous practise in a variety of ray diagrams pertaining to Total Internal Reflection (TIR) / critical angle to build the confidence of the students. - Emphasise in the discussion that the angle of incidence is with the normal and not with the reflecting surface. - Teach all the topics related to the concept 'Light' with relevant diagrams and make sure to mark the angles in the diagram.
(iii) (b) 2.	- Most candidates erroneously wrote: 90^0, 47^0. This was an unexpected question, which was inclined towards geometry. - Majority of the candidates wrote the answer as 45^0 thinking that it was an isosceles triangle. - Some candidates calculated all three angles but did not mention which was the angle of the prism.	- Explain the geometrical concepts involved in these phenomena properly. - Develop the habit of explaining concepts by drawing diagrams, wherever possible. - Encourage the students to calculate angles each time, the ray strikes the surface in a denser medium by using laws of refraction and basic geometry.
(iii) (b) 3.	- A large number of the candidates mentioned any colour without any idea of the concept. - Many candidates were found to be confused by this question. - Since the question was challenging, the majority of the candidates were unable to attempt. - Most candidates answered 'red' as the relation between critical angle and wavelength which shows the lack of clarity in the understanding of the concept.	- Explain the relation between the critical angle and the wavelength to the students. - For different colours i.e. VIBGYOR, the different physical quantities change in the following manner. (V:Violet, I:indigo, B:blue, G:green, Y:yellow, O:orange, R:red) ➤ $\mu_V > \mu_I > \dots > \mu_R$ μ:Refractive index. ➤ $f_V > f_I > \dots > f_R$ f:frequency. ➤ $\lambda_V < \lambda_I < \dots < \lambda_R$ λ:Wavelength. ➤ $L_V > L_I > \dots > L_R$ L:Lateral displacement. ➤ $S_V < S_I < \dots < S_R$ S: Speed of light. ➤ $\delta_V > \delta_I > \dots > \delta_R$ δ:Angle of deviation. ➤ $C_V < C_I < \dots < C_R$ C:Critical angle. dependence of these quantities should be explained clearly.

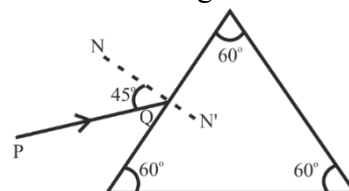
MARKING SCHEME

Question 4

(i)	$U = -36 \text{ cm}, V = 72 \text{ cm}$ $\therefore \frac{1}{f} = \frac{1}{72} - \frac{1}{-36}$ Or $\frac{1}{f} = \frac{1}{72} + \frac{1}{36}$ [with correct sign convention] $\therefore \frac{1}{f} = \frac{1+2}{72} = \frac{3}{72} \quad \therefore f = 24 \text{ cm}$ $P = \frac{100}{24} = \frac{25}{6} = +4.16 \text{ or } +4.17 \text{ D or } 4.1 \text{ D or } 4.2 \text{ D}$
(ii)	(a) micro-waves / television waves. (b) X-ray (c) Travel with speed $3 \times 10^8 \text{ ms}^{-1}$ in vacuum and air or do not require a medium to travel or can be reflected or can be refracted/ Scatters least. (<i>Any one correct property.</i>)
(iii)	a. (Red colour has) longer wavelength/ (and therefore they) scatter the least/can be seen from a longer distance/travel long distance through the fog. b. 1. $(C = 133 - 90) = 43^{(0)}$ 2. $(A = 90 - (180 - 133) = 90 - 47 =) 43^{(0)}$ 3. Violet or indigo

Question 5

- (i) (a) Refractive index of glass with respect to water is $\frac{9}{8}$. [3]
 Find the refractive index of water with respect to glass.
- (b) Name the principle used to find the value in part (a).
- (c) If we change the temperature of water, then will the ratio $\frac{9}{8}$ remain the same? Write **Yes** or **No**.
- (ii) Light travels a distance of ' $10x$ ' units in time ' t_1 ' in vacuum and it travels a distance of ' x ' units in time ' t_2 ' in a denser medium. Using this information answer the question that follows: [3]
- (a) 'Light covers a distance of ' $20x$ ' units in time ' t_1 ' in diamond.' State true or false.
- (b) Calculate the refractive index of the medium in terms of ' t_1 ' and ' t_2 '.
- (iii) A monochromatic ray of light is incident on an equilateral prism placed at **minimum deviation position** with an angle of incidence 45° as shown in the diagram. [4]
- (a) Copy the diagram and complete the path of the ray PQ.
- (b) State *two* factors on which the angle of deviation depends.

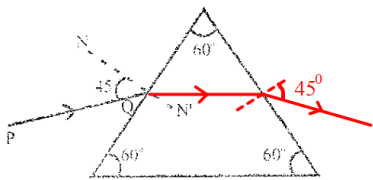


Comments of Examiners		Suggestions for teachers
(i) (a)	Most of the candidates answered the question correctly as they provided the answer in the fractional form.	Ensure frequent practise for these types of questions.
(i) (b)	- Majority of the candidates attempted this question correctly. - The law of refraction /reflection/Snell's law etc. was written instead of 'the principle of reversibility'.	- Explain the principle of reversibility. - Ask students to focus on the words used in the question such as the 'Name' means that only the name to be mentioned and 'State' means the elongated form that is the statement as stated in the law or principle.
(i) (c)	- Many candidates answered this question correctly. - Some candidates wrote the answer without a conceptual understanding. - Few of the candidates were unable to express their ideas correctly as 'Yes' and 'No'.	Train students to answer the question that is being put forward in the form of 'Yes or No', with a clear response.
(ii) (a)	- Many candidates were unable to comprehend this question correctly. - A large number of candidates erroneously answered this question as they were unable to connect the fact that light travels fastest in a vacuum. So instead of 'false', the candidates mentioned 'true'.	Explain to the students that the 'Light travels fastest in a vacuum and hence covers the maximum distance in vacuum than in any other medium' in detail.
(ii) (b)	- Many candidates were not able to solve this question correctly as it reflects the lack of clarity in the understanding of the concepts. - Some candidates were able to calculate velocity separately but made calculation errors in finding the refractive index. - Instead of calculating the velocities in both the media, candidates used $20X$ for calculation.	- Guide students on how to find the refractive index from the velocities. - Give unfamiliar situations and show the way to employ the formula, assess the information provided in the question, and attempt to determine the unknown by utilising the given data. - Provide rigorous practice of application-based questions by introducing them in the teaching-learning process.
(iii) (a)	- Many candidates answered this question correctly, but very few missed the arrows of the light ray. - Few candidates did not draw the refracted ray parallel to the base and	Regular practice of ray diagrams ensuring arrows are always marked will help eliminate oversights. Also, deal effectively in class the conditions for minimum deviation and how to incorporate them in a diagram.

	some candidates did not mark the $\angle e = \angle i$. Some candidates showed the total internal reflection of the ray and other candidates drew dotted lines instead of continuous lines.	Explain to the students that during minimum deviation $\angle i$ is always equal to $\angle e$ but refracted ray is parallel to the base only in the case of equilateral prism or isosceles prism.
(iii) (b)	- Majority of the candidates wrote more than two factors affecting the angle of deviation. - Some candidates were confused with the lateral displacement and wrote the factor as thickness. - The common errors found were: - The colour/wavelength of the prism. - The angle of incident ray. - The refractive index of light.	- Review and demonstrate all factors influencing each physical quantity covered in the chapter and their impact repeatedly. Through frequent repetition, students will better retain this information. - Make students aware that it is important to avoid using the wrong terms from the beginning of the concept in the class.

MARKING SCHEME

Question 5

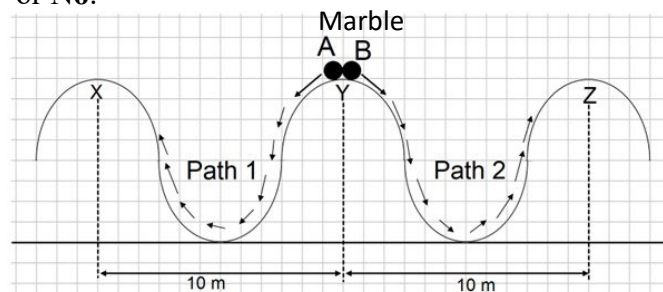
(i)	(a) ${}_g\mu_w = \frac{1}{{}_w\mu_g} = \frac{1}{\frac{8}{9}} = \frac{9}{8}$ or 0.88 or 0.8 or 0.89 or 0.9. (b) (Principle of) reversibility (of light.) (c) No/ it will decrease/or implied.	
(ii)	(a) False/F/or implied. (b) $\mu = \frac{\frac{10x}{t_1}}{\frac{x}{t_2}}$ or $\mu = \frac{10x}{t_1} \times \frac{t_2}{x} - 1$ (any one velocity is calculated.) $\mu = \frac{10t_2}{t_1}$	
(iii)	(a) 	Showing refracted ray parallel to the base. Completing the path of emergent ray bending away from the normal. Avoid missing arrows on refracted or emergent ray.

- | | |
|-------|--|
| (iii) | (b) Angle of deviation depends on: <ul style="list-style-type: none"> the angle of the prism the refractive index of the medium/material of the prism. Wavelength/colour of light, the angle of incidence. (<i>any two points, conditions apply.</i>) |
|-------|--|

Question 6

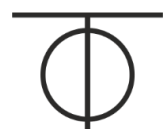
- (i) (a) Define Centre of Gravity. [3]
- (b) A hollow ice cream **cone** has height 6 cm.
1. Where is the position of its centre of gravity from the **broad base**?
 2. Will its position change when it is filled completely with honey? Write **Yes** or **No**.

- (ii) [3]



Two identical marbles A and B are rolled down along Path 1 and Path 2 respectively. Path 1 is **frictionless** and Path 2 is **rough**.

- (a) Which marble will **surely** reach the next peak?
 - (b) Along which path/s the **mechanical energy** will be conserved?
 - (c) Along which path/s is the law of **conservation of energy** obeyed?
- (iii) Given are two pulleys. [4]
- (a) Copy and complete the labelled diagram connecting the two pulleys with a tackle to obtain Velocity Ratio = 2.
 - (b) If Load = 48 kgf and efficiency is 80% then calculate:
 1. Mechanical Advantage.
 2. Effort needed to lift the load.



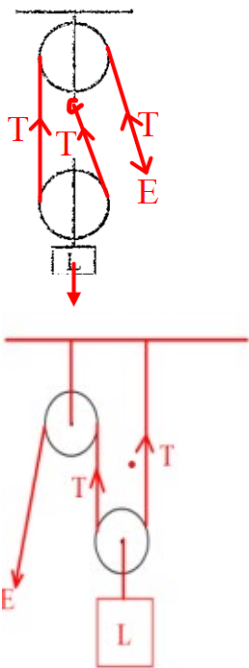
Comments of Examiners	Suggestions for teachers
(i) (a) - Many candidates were unable to define the centre of gravity correctly, thus pointing towards a lack of understanding of the concept of moment produced by the weights of the molecules. - Majority of the candidates were confused with the definition of the centre of gravity and provided incomplete definitions. The missing keywords found were 'point', 'moment of weights', and 'sum of moments is zero'. Many candidates wrote that the sum of weights/masses of particles constituting the body is 'zero'.	- Ensure that the students understand the concept clearly. - Discuss in class to build an understanding of the concept of moments produced by molecular weights using the diagrammatic representation of the line of action of molecular weight and its moment arm. - Though definitions need to be verbatim, emphasis must be laid on the fact that all the key terms are present so that the meaning/essence is not lost. Also, bring the attention of the students to the language while framing the answer. - Reinforce the correct understanding by presenting incorrect definitions in class and challenge students to identify the errors. - Discuss in a way that the students form a mental picture while learning definitions in the classroom.
(i) (b) 1. - Many candidates answered it erroneously as they wrote $h/3$ without substituting 'h' due to oversight or lack of conceptual understanding. - Some candidates erroneously obtained answers as: $6/4=1.5\text{cm}$ $6/2=3\text{ cm}$ - While identifying the centre of gravity of various objects, candidates should prioritise using precise terminology instead of general phrases like "at the centre," which often led to loss of marks. - Some candidates calculated 4 cm but were unable to mention from the top. - Very few candidates were able to answer this question correctly.	- More practise is required of application-level questions instead of direct questions. - Lay special emphasis to the centre of gravity of all regular bodies. - Discuss in class that in case the data / values are mentioned in the question, then the expected answer must be provided in a value/number and not the formula. The question is equivalent to a numerical. - Instruct students even if the basic question remains the same, make changes in the answer as per the given information.
(i) (b) 2. - Most candidates answered this part of the question correctly. - Some candidates were found to be unclear on the concept 'Centre of gravity depends on mass distribution'.	- Discuss in class with different examples that the centre of gravity depends on the distribution of mass. - Make a point during the discussion that in uniform solids such as cube, cuboid, cylinder, whether the solid is hollow or filled distribution of mass remains uniform but in the case of a

		cone, more mass will be towards the base so the position of the centre of gravity will change.
(ii) (a)	- Considering the question in totality, very few candidates could comprehend the question. - No clarity of the concept of frictionless and rough surfaces. - Confusion was found to be in comprehending the diagram. - Some candidates were not clear that friction opposes motion.	- Give more practice with application-based questions so that students can apply the concept in their daily lives. - Explain to students the meaning of ideal conditions. - Reinforce to the students that 'Friction causes dissipation of energy'.
(ii) (b) and (ii) (c)	- Majority of the candidates answered this question correctly. - Some candidates lost marks here as they failed to recognise that mechanical energy is only conserved, when there is no friction. - Some candidates could not differentiate between the law of conservation of mechanical energy and the law of conservation of energy and the condition under which they are true. So, for many candidates all the answers went wrong. - Candidates provided the answer in terms of right/left side, A/B instead of path 1 and path 2. - Many candidates attempted these parts by guesswork indicating lack of clarity in the concepts.	- Discuss in the class the different forms of energy, their interconversion, and the law of conservation of energy as a whole and not compartmentalised. - Lay emphasis on the critical distinction between mechanical energy conservation, which requires a frictionless environment, and total energy conservation, which applies universally regardless of friction.
(iii) (a)	- Most candidates connected the 'tackle' correctly. However, the missing pieces of the diagram were: - The tension direction was not properly indicated. - Few candidates marked tension in the sections of the tackle as T_1, T_2, T_3. - The direction of effort was not indicated. - A bulging tackle. - No rigid support was shown. - Few candidates connected the 'tackle' in such a way that the proper logic was not visible in the answer.	- Discuss the intricacies in drawing the diagram in the class. - Give adequate practice by frequently asking this type of question in the class. - Make students aware of the common errors that they make while drawing these diagrams. - Replicate the diagram exactly as presented in the question, avoiding any unnecessary alterations. - Ask students to mark the tension, effort, and load, regardless of whether explicitly requested in the question.
(iii) (a) 2.	- Most of the candidates attempted this numerical correctly. - Though the substitution was correct, many candidates kept the answer as	Advise students to answer all numerical in decimal form up to two decimal places unless it is a 'compare' question.

	a fraction, $\frac{8}{5}$ and did not convert it to a decimal, 1.6. Some candidates write the unit for Mechanical advantage which was not required.	In a compare question, tell students to answer in the ratio form.
(iii) (b) 2.	- Most of the candidates answered it correctly. - Some of the errors observed were: - carried out calculation by using ideal MA. - Calculation error. - Units were missing in some cases and confusion was seen in kgf and N.	- Give adequate practice of numerical to the students. - Emphasise correct answers with the unit. - Tell students to recheck answers of numerical and calculations.

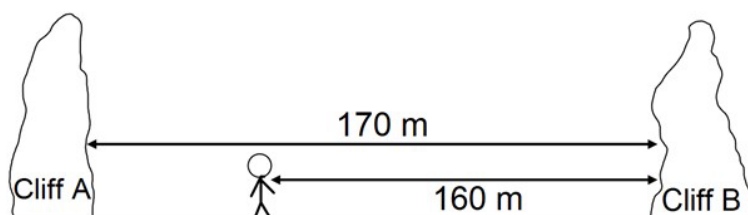
MARKING SCHEME

Question 6

(i)	(a) It is a <u>point</u> about which the (algebraic) sum of the moments of the molecular weights of the body is zero or implied/ It is a point through which the weight of the body acts (vertically downwards or towards the centre of the earth.) or implied. (b) 1. $\frac{6}{3} = 2$ cm above the broad base 2. Yes	
(ii)	(a) Marble A/ A or implied. (b) Path 1/1/ marble A/A/ or Implied. (c) Path 1 as well as Path 2/ both paths/1 and 2/A and B or implied.	
(iii)	(a) Correct connection of tackle. Labelling T and E correctly with directions. (b) 1. $Efficiency = \frac{M.A.}{V.R.} \therefore \frac{80}{100} = \frac{M.A.}{2}$ $\therefore M.A. = 1.6$ [Substitution is a must] 2. $M.A. = \frac{L}{E} \therefore 1.6 = \frac{48}{E}$ [Substitution in any form] $\therefore E = \frac{480}{16} = 30$ kgf /300N/294 N	

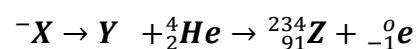
Question 7

- (i) (a) Name the waves used in SONAR. [3]
 (b)

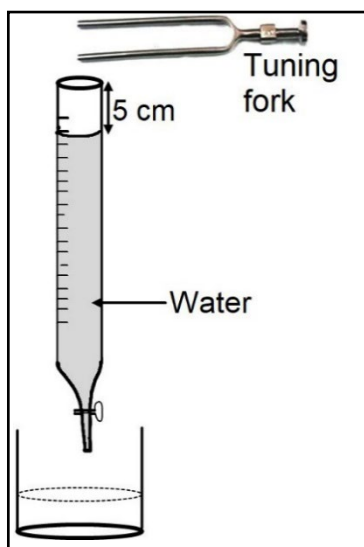


In the above diagram Lata stands between two cliffs and claps her hands. Determine the time taken by her to hear the **first** echo.
 Speed of sound in air 320 ms^{-1} .

- (ii) (a) Complete the following radioactive reaction: [3]



- (b) Uranium is available in two forms U-235 and U-238. Which of the two isotopes of Uranium is **more** fissionable?
 (iii) In the given diagram, a vibrating tuning fork is kept near the mouth of a burette filled with water. The length of the air column is adjusted by opening the tap of the burette. [4]
 At a length of 5 cm of the air column, a **loud sound** is heard.



- (a) Name the phenomenon illustrated by the above experiment.
 (b) Why is a loud sound heard at this particular length?
 (c) If the present tuning fork is replaced with a tuning fork of higher frequency, should the length of the air column increase or decrease to produce a **loud sound**? Give a reason.

(i) (a)	- Majority of the candidates attempted this part of the question correctly. - Some of the candidates had written 'ultraviolet/infrared/radio waves' instead of 'ultrasonic waves'.	- Discuss with the students and make them to recall the three types of mechanical waves (electromagnetic spectrum): ultrasonic, infrasonic, and sonic waves. - Explain the distinctions between electromagnetic and mechanical waves to foster students' understanding.
(i) (b)	Since the question was thought-provoking, many candidates solved the numerical by erroneously considering the distance to be 10 m, not realising that a minimum distance of 17 m is required to hear an echo in the air.	- Give sufficient practice of numerical involving principle of echo. - Discuss with the students in detail to build a comprehensive understanding of the concept of echo. - Explain the conditions of echo to the students in detail. - Tell students to be mindful of possible errors that may occur while measuring the distance of an echo.
(ii) (a)	- Most of the candidates were found to be confused due to the presentation of the nuclear equation. - Many candidates were not able to comprehend the equation. - Reverse calculation was difficult for many candidates, when more than one radioactive emission happened.	- Explain the nuclear reaction concept in detail. - Discuss in class that the emitted particle or radiation cannot become reactant for the next step of emission. - Write the radioactive reaction in two separate steps to avoid confusion among students.
(ii) (b)	- Majority of the candidates answered it correctly, however quite a number of candidates erroneously answered it as U_{238} instead of U_{235}. - Some candidates had erroneously written both the names.	Lay emphasise while explaining in the class on the fact that there are two isotopes of uranium and out of the two U_{235} is more fissionable.
(iii) (a)&(b)	- Majority of the candidates answered it correctly as 'resonance' barring few of the candidates who erroneously wrote 'forced vibrations'. - The terms 'natural', 'matching' and 'amplitude' were missing in the explanation of resonance. - Some of the candidates were confused with frequency and length of air column.	- Tell students to mention all keywords while writing definitions. - Reinforce in the class that the length of vibrating air column is responsible for resonance. - Discuss with the students to link one point to the next and come to the conclusion covering all points. For e.g. Forced freq. of tuning fork matches natural freq. of air column

	- Many candidates answered in terms of 'frequency of the burette' rather than the 'air column'. - A large number of the candidates did not mention that more the amplitude louder the sound, for this part of the question.	→ increases amplitude of vibration of air particles → increases loudness. Demonstrate the concepts practically in the class.
(iii) (c)	- Most candidates answered 'decreased' correctly. However, many of the candidate missed writing the reasoning or not correct reasoning mentioned. This indicated that the relation between length of air column and frequency was not clear to the candidates. - Some candidates mentioned that frequency of vibrating air column is inversely proportional to length of air column but still gave the answer as increases which indicates the lack of clarity of the concept. - Few of the candidates wrote 'length of water column' instead of 'air column'. - Instead of 'frequency of tuning fork', 'frequency of burette' was written.	- Explain the concept of factors affecting the frequency to the students and also explain how it affects. - Give rigorous practice of application-level questions based on this concept. - Provide more opportunities to the students to demonstrate a deeper understanding and application of concepts.

MARKING SCHEME

Question 7

(i)	(a) Ultrasonic (waves)/ultrasound (b). $v = \frac{2d}{t} \therefore 320 = \frac{2 \times 160}{t} \therefore t = \frac{320}{320} = 1 \text{ s}$ If d=10 m is calculated and identified d<17 m and no echo is heard.
(ii)	(a) ${}^{238}_{92}\text{X} \rightarrow {}^{234}_{90}\text{Y} + {}^4_2\text{He} \rightarrow {}^{234}_{91}\text{Z} + {}^0_{-1}\text{e}$ (b) U-235
(iii)	(a) Resonance (b) <u>Natural</u> frequency of the tuning fork <u>matched</u> with the <u>natural</u> frequency of the air column and the <u>amplitude</u> of vibrations of air particles increased/Generalised statement can be accepted (Natural is must at any one place, frequency matching and increased amplitude is a must.) (c) Decrease. The frequency of vibrating air column is inversely proportional to the length of the air-column/ $f \propto 1/t$.

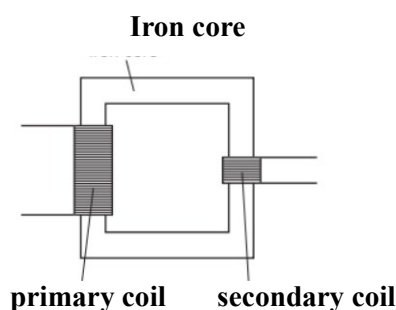
Question 8

- (i) The voltage - current readings of a certain material are shown in the table given below: [3]

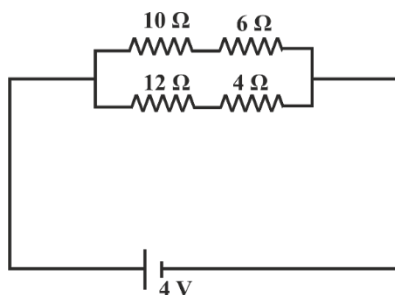
Voltage (V)	10 V	20 V	30 V
Current (I)	2 A	3 A	4 A

Study the table.

- State whether the conductor used is ohmic or non-ohmic.
 - Justify your answer.
 - State Ohm's law.
- (ii) Below is the diagram of a transformer: [3]



- Identify the type of transformer.
 - In this type of transformer which of the wire is **thicker**, the primary or the secondary? Give a reason.
- (iii) Study the diagram: [4]



- Calculate the total resistance of the circuit.
- Calculate the current drawn from the cell.
- State whether the current through $10\ \Omega$ resistor is greater than, less than or equal to the current through the $12\ \Omega$ resistor.

Comments of Examiners		Suggestions for teachers
(i) (a)	Many candidates were found to be confused in this question.	To decide whether it is an ohmic or non-ohmic conductor, it is crucial to

	Despite this question being awarded free marks, many candidates made the error of determining the answer solely by examining the slope. They overlooked the crucial detail that the V against I graph must pass through the origin.	consider not just the slope but also ensure that the V - I graph passes through the origin, even if it's not explicitly marked. Make them experimentally verify Ohm's law and plot a V - I graph using the data collected.
(i) (b)	- Many candidates interpreted this question erroneously. - Those candidates who only calculated the slope wrote it as Ohmic conductor. - Some candidates who calculated V:I ratio wrote it as non-ohmic conductor. - The candidates who calculated both were unable to provide an explanation for their answer in (a).	- Ask students to verify, even if the slope is constant, whether the straight line passes through the origin. - Reinforce the concept by providing more questions of this type in the class.
(i) (c)	- Most of the candidates wrote answers correctly. - Few candidates did not mention the constant temperature in their answer. - Some candidates lost marks as only mathematical relation based on Ohm's law was given instead of a proper statement. - Several candidates wrote 'I is directly proportional to R'. - A very few candidates wrote 'I is inversely proportional to V'.	- Teach statements and laws with the proper terms and conditions. - Give explanation of laws and statements to the students. - Ask students to articulate complete statements when defining Ohm's Law as these concepts are prone to misunderstanding when only partially explained.
(ii) (a)	Some candidates were confused with the wiring in the diagram and hence misinterpreted as step-up transformer.	- Explain the students the transformer with schematic diagram. - Lay emphasis while teaching emphasise the distinction between step-up and step-down transformers. - Stress while teaching this concept mathematical relation relating n, E, I can be stressed upon. - Show video presentation on types of transformers.
(ii) (b)	Many candidates wrote an incorrect answer for this question	Highlight in class the disparity between step-up and step-down

	implying was a lack of understanding of the concept. - A large number of candidates misinterpreted the number of turns in the diagram as indicative of the thickness of the wire, leading them to incorrectly identify the primary coil as thicker. Instead of directly stating that the secondary wire carries more current and was therefore thicker, they attempted to explain in terms of voltage, resulting in the loss of marks. - Basis conceptual things like more current needs less the resistance was not known to many candidates.	transformers based on voltage, current, number of turns, and thickness. As power remains the same, introducing the formula $V_S / V_P = N_S / N_P = I_P / I_S$ assists students in discerning the correct relationship among these parameters.
(iii) (a)	- Many candidates solved this numerical correctly. - Few candidates were not able to write the units correctly. - Some candidates did the numerical correctly but lost marks due to mathematical incorrect step. e.g $1/16 + 1/16 = 2/16 = \underline{1/8} = 8 \Omega$ - Some candidates lost marks due to incorrect formula or wrong substitution.	- Sufficient practice of numerical involving combination of resistors to be given. - Insist on writing correct answers with unit. - Give practice of solving numericals in class.
(iii) (b)	- Most candidates answered this question appropriately. - A few candidates missed mentioning the unit. - Error in the first part has resulted in the loss of this mark in the case of some candidates.	- Lay emphasis on the importance of including units in numerical answers, stating clearly that marks will not be awarded for the numerical portion, if units are missing. - Give sufficient practice of numerical.
(iii) (c)	- Most candidates went to tedious calculations ending up with the erroneous answer. - The candidates did not realise the fact that the current splits equally as the resistance of both the branches is the same. - How to find current in a branch of combination of resistors was not clear to many candidates.	- Explain the concept of series and parallel combination in solving this type of numerical. - Provide adequate practice of numerical. - Elucidate in class that 'In a parallel circuit the current was inversely proportional to the total resistance of that branch'.

	Some candidates substituted $V = 4V$, $R = 10\Omega$ and calculated the answer.	
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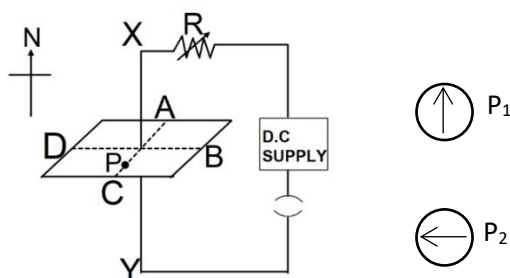
MARKING SCHEME

Question 8

(i)	(a) Non-ohmic (b) Even though slope constant $V:I$ ratio is not constant. $\therefore$ The graph line will not pass through the origin. Graph (line) will not pass through the origin/ Resistance is not constant/ diagram VI graph not passing through origin/ V/I not constant or implied. (c) Provided the physical condition/temperature of the conductor remains the same, current through the conductor is directly proportional to the potential difference across the ends of the conductor.
(ii)	(a) Step-down transformer (b) Secondary As the current in the secondary coil increases/ to decrease the resistance/to minimise the loss of energy/ to prevent overheating of coil/ copper loss or implied.
(iii)	(a) $R = \frac{16 \times 16}{16 + 16}$ [Substitution in any form] $\therefore R = 8\Omega$ (b) $V = IR \therefore 4 = I \times 8 \therefore I = \frac{4}{8} = \frac{1}{2} = 0.5 A$ Current through both the resistors is the same.

Question 9

- (i) 85 g of water at 30°C is cooled to 5°C by adding certain mass of ice. Find the mass of ice required. [3]
[Specific heat capacity of water = $4.2 \text{ Jg}^{-1}\text{C}^{-1}$, Specific latent heat of fusion = 336 Jg^{-1}]
- (ii) (a) Why does it become pleasantly warm when the lakes start freezing? [3]
(b) Water freezes to form ice. What change would you expect in the average kinetic energy of the molecules?
(c) Which will contain more heat energy 1 g of ice at 0°C or 1 g water at 0°C ?
- (iii) (a) State one factor that affects the magnitude of induced current in an AC generator. [4]
(b) Given below is a circuit to study the magnetic effect of electric current. ABCD is a cardboard kept perpendicular to the conductor XY. A magnetic compass is placed at the point P of the cardboard. P₁ and P₂ are the positions of the magnetic compass, before and after passing a current through XY respectively.



1. Name the **rule** that is used to predict the direction of deflection of the magnetic compass.
2. State the direction of current in the conductor (X to Y or Y to X) when the circuit is complete.
3. If resistance R is increased, then what will be the effect on the magnetic lines of force around the conductor?

Comments of Examiners		Suggestions for teachers
(i)	• Most of the candidates did not attempt this question. • Substitution was not done correctly in the numerical. Specially for melting of ice only ml was considered and not $mc\Delta t$ for increase in temperature of water. • There were calculation errors. • Error was made in writing the units.	- Give more practice of numerical involving different types. - Insist upon answers with correct unit. - Advise to students to construct an equation of this type of numerical using heating curve. This will enable correct analysis of the numerical. - Reinforce in students that in order to simplify the calculation, that 4.2 is a factor of 336 which can be taken outside the bracket.
(ii) (a)	• In this question, candidates got confused with release of heat energy to the atmosphere during freezing and absorption of heat energy during its melting. • Most candidates missed out mentioning '<u>large</u>' amount of heat. • Few candidates mentioned specific heat capacity. • Many candidates wrote vague and incorrect answers. • Numerous candidates lost marks on this question due to attempts to abbreviate their responses, consequently omitting crucial keywords such as 'releasing latent heat of fusion'.	- Explain that the volume of heat released during change of phase and during normal cooling. - Elucidate in class that heat is released during freezing and absorbed during melting. - Underline keywords in reasoning questions. This practice ensures alignment between the number of keywords identified and the corresponding marks allotted for the question.

(ii) (b)	- As there was no concept clarity for conditions during change of state, most of the candidates answered increases or decreases. - Majority of the candidates did not recall that the kinetic energy of molecules remains constant during a phase change.	- Explain in detail that what happens to the kinetic energy and potential energy during change of phase with examples. - Address the fact that temperature and average kinetic energy remains constant during change of state from grade VIII. - Lay emphasis on the energy changes that occur at various stages of a phase-change graph and also that heat absorption did not always result in increase of temperature.
(ii) (c)	- Most of the candidates answered this question correctly. - Very few candidates answered as ice due to careless error or due to lack of clarity of the concept.	- Clarify the misconception about the relationship between temperature and heat energy. Make a point that when both water and ice are at the same temperature (0°C), the heat energy content of water is higher than that of ice at this temperature. - Discuss explicitly latent heat concept in class.
(iii) (a)	- Many candidates answered this question correctly. - In this question, many of the candidates were confused with factors affecting speed of rotation of a d.c motor. - Some candidates gave more than required factors and all were not correct, so they lost marks. - Some candidates did not attempt this question.	- Explain a.c generator with the diagram in detail. - Lay emphasis on the difference between d.c motor and ac. generator. - Explain the factors affecting the induced emf in detail. - Ensure that students provide the exact number of points requested in the question.
(iii) (b) 1.	- A large number gave incorrect answers indicating lack of conceptual clarity. The errors were as follows: ➤ Fleming's left-hand rule. ➤ Fleming's right-hand rule. ➤ Ampere's right-hand rule. ➤ Clock rule.	- Emphasise the application of these rules and clarify which physical quantity each rule helps to identify. - Give practice of these kinds of questions by changing the orientation of the diagram.
(iii) (b) 2.	- Even though question was direct and simple, candidates found it difficult to comprehend the diagram. - Guesswork was found to be done by many candidates.	Although one can use the right-hand grip rule, it becomes simpler to remember that when the current flows through the conductor, magnetic needle deflects in the direction of the magnetic field.

		- Give more practice of these kinds of questions to the students.
(iii) (b) 3.	- Many candidates answered it correctly. - Also, because of confusion in the previous part, the candidates made errors in this. - Most of the candidates were unable to understand the concept correctly.	- Explain how change in resistance affects the current and thereby the effect on the strength of the magnetic field. - Give practical demonstration of the nature of the magnetic field lines on varying the current.

MARKING SCHEME

Question 9

(i)	(By principle of mixtures Heat lost = heat gained $mc\Delta t_{\text{water}} = (ml + mc\Delta t)_{\text{ice}}$ $\therefore 85 \times 4.2 \times (30 - 5) = m \times 336 + m \times 4.2 \times 5$ $\therefore 85 \times 4.2 \times 25 = 4.2 \times m(80 + 5)$ $\therefore 85 \times 25 = m \times 85 \therefore m = 25 \text{ g}$
(ii)	a. Large quantity of heat/ latent heat of fusion/ latent heat/ 336 J/g or value or value in other units/ is given out/ liberated/ released (in the atmosphere) or implied. b. (There is) no change (in the average kinetic energy of the molecules.) or implied. c. 1 g of water at 0 °C (contains more heat energy than 1 g of ice at 0 °C).
(iii)	a. Induced current depends on is directly proportional to - The strength of the magnetic field - The number of turns of the coil. - The area of the coil. - The speed of rotation of the coil. - Rate of change of magnetic flux. <i>[Any one factor.]</i> b. 1. Right hand thumb rule/ Amphere's swimming rule/maxwell's cork screw rule/SNOW rule/Right-hand grip rule/ Right hand palm rule. 2. x to y 3. The number of lines of force will decrease/ less crowded/ spaced out/ less dense or implied.

Note: For questions having more than one correct answer/solution, alternate correct answers/solutions, apart from those given in the marking scheme, have also been accepted.

GENERAL COMMENTS

Topics found difficult by candidates

Mechanics

- **Question 2 (iii):** Numerical on moments based on non-uniform distribution of mass.
- **Question 6 (ii)(b) & (ii)(c):** Law of conservation of energy vs. conservation of mechanical energy.
- **Question 6 (i):** Definition of centre of gravity.
- **Question 6 (iii):** Modification in the diagram of block and tackle system.

Light

- **Question 4 (i):** Sign convention in lenses.
- **Question 4 (iii):** Diagram based questions from prism.
- **Question 4 (iii)(b) 1:** Calculation of critical angle.

Sound

- **Question 1 (iv):** Resonance definition.
- **Question 7 (i) (b):** Numerical on principle of echo.
- **Question 7 (iii):** Relation of length of air column with frequency.

Electricity

- **Question 1 (x):** Difference between voltage drop and terminal voltage.
- **Question 2 (iv):** Relationship between resistance, current and brightness of a bulb.

Calorimetry

- **Question 9 (ii) (a & b):** Energy absorption and release during change of phase.
- **Question 9 (ii) (c):** Heat numerical.

Electromagnetism

- **Question 9 (iii)(b) 1:** Conditions under which different rules such as Right-hand thumb. rule, Fleming's right-hand and left-hand rule etc.
- **Question 9 (iii) (a):** Electric motor and AC generator.

Radioactivity

- **Question 1 (iv):** What happens inside nucleus during beta emission.

In general:

- Experimental questions.
- Graph-based questions.
- Units of physical quantities.

Concepts in which candidates got confused

Mechanics

- Numerical using principle of moment for non-uniform distribution of mass.
- Displacement in calculating work when diagrammatic information is given.
- Purpose of one more pulley in the movable block of block and tackle system.
- Proper definition of the Centre of Gravity and its position in different shapes.
- Difference between principle of conservation of energy and principle of conservation of mechanical energy.
- Modification in the diagram of block and tackle system as per the given diagram.

Light

- Magnification for different cases of lens diagrams.
- Application of combined refraction between lenses and glass block.
- Sign convention in lens formula.
- Diagram based questions from optics.
- Solving numerical based on $\mu = \frac{\text{Light Speed}_{\text{air}}}{\text{Light speed}_{\text{medium}}}$ involving non-numerical data.

Sound

- Examples of different vibrations.
- Time to hear echo for humans remain the same in different media but the distance changes in different media.
- Frequency is inversely proportional to length.

Electricity

- Diagrammatic questions on household wiring.
- Ohm's law diagram and condition to identify Ohmic conductor.
- Calculation of current in parallel branches.
- Relation between resistance, current drawn and thereby energy consumed.

Calorimetry

- Energy absorption and release during melting and freezing.
- Numerical involving mixtures and change of phase of the substances mixed.

Electromagnetism

- Confusion in factors affecting the Speed of rotation in DC motor and induced current in AC generator.
- Confusion in comprehending the diagram when orientation of the diagram is changed.

Radioactivity

- Changes inside nucleus during beta emission.
- Completion of balancing radioactive reactions when the reactions are given sequentially and not separate.

Suggestions for candidates

- Use the reading time 15 minutes judiciously to make the proper choice of the questions from the relevant sections by reading all the parts and requirements of the questions carefully. Also, use the same data mentioned in the question.
- Focus on the key words in the answer statement.
- Make sure not to leave any topic as in certain sections, all questions come compulsory.
- Avoid changing the order of sequence of questions and numbering system.
- Make sure that the handwriting is neat and legible.
- Learn principles, laws and definitions accurately.
- Practice Ray diagrams and other diagrams periodically.
- While solving a numerical, it is advisable that the formula needs to be written in the beginning. Show essential steps in the numerical solutions and be mindful while substituting values, ensuring that they are in the same unit.
- Make sure to write unit with every physical quantity.
- Be regular in self-study. Complete the syllabus well in time. Thorough revision of all topics is required.
- Solve the Competency Focused practice questions, previous board papers and specimen paper issued by the CISCE.
- Give more emphasis on writing rather than memorising.
- Inculcate the habit of underlining the important points or keywords in the answer.
- Practice solving a variety of physics problems from textbook or other resources to reinforce understanding of concepts and familiarise with the type of questions that may be asked for the exams.
- Try to apply the concepts (related to physics) with the real-life examples.
- Never leave any answer in fractions and use the fractional value calculated before in the next part of the calculation in numerical.
- Recheck all answers prior to submitting the paper.
- Avoid including any additional incorrect points as that may result in marks being deducted.
- Watch e-videos on the concepts and try to make some projects/models that work on the principles learnt in the class.